

Large-aperture Optical Antenna with Vertical Emission in Silicon-on-Insulator

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Abstract *Vertically emitting optical antennas with large mode-fields diameters are key for free space optics and atomic spectroscopy, but have not been yet reported in silicon-on-insulator. We present a novel vertically radiating antenna with a mode-field diameter of $200\ \mu\text{m}$ by $320\ \mu\text{m}$ with measured 48 % efficiency. ©2026 The Author(s)*

Introduction

The rapid expansion of silicon photonics is fuelled by its capacity to integrate intricate optical circuits into compact, mass-producible chips using established CMOS manufacturing techniques [1]. While this technology has matured significantly within the telecommunications and data center sectors, adapting it for specialized applications, such as free space optics [2, 3], atomic spectroscopy [4] and the development of ultra-stable laser resonators [5], presents a formidable technical challenge. These advanced applications require antennas that project a beam with mode field diameters (MFD) exceeding $100\ \mu\text{m}$ and perfect vertical radiation to reduce packaging costs [2] and simplify alignment with other components [4]. These two requirements represent a massive difference compared to the $10\ \mu\text{m}$ modes standardly used for vertically interfacing with optical fibers [6, 7].

Achieving the sizes of the beams required for these applications requires an antenna grating with an exceptionally low radiation strength. In the silicon-on-insulator (SOI) platform, these antennas are scarce and do not allow vertical radiation [3]. Existing progress in large-beam antennas with perfectly vertical emission has been made on silicon nitride (SiN) platforms [2, 4], where the smaller index contrast compared to SOI naturally produces weaker gratings. These results leveraged inverse design to create resonating grating structures [2] or polymer top-layers that alter the radiation angle of non-vertical emission antennas [4], resulting in projected vertical beams with mode field diameters (MFD) of $150\ \mu\text{m} \times 150\ \mu\text{m}$.

In this work, we present what is, to the best of our knowledge, the first large-aperture antenna in SOI with $220\ \text{nm}$ core thickness to achieve vertical emission, which is also the largest vertically emitting antenna reported in the literature.

To achieve the necessary low radiation strength without compromising verticality, we utilize a long dual-period unit cell (inset of Scanning Electron Microscope (SEM) in Fig. 1(a)), which allows the

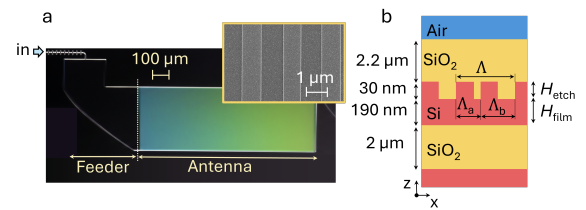


Fig. 1: (a) Fabricated vertical antenna with parabolic feeder. Inset of SEM with period shape. (b) 2-D cut of SOI platform employed in this work showing the periodic perturbation.

radiation of a high-order Floquet harmonic, specifically the sixth-order harmonic.

We fabricated and characterized this large-aperture radiator, which operates with transverse-electric (TE) polarization at $\lambda = 1.55\ \mu\text{m}$. Measurements show it vertically projects a MFD of $200\ \mu\text{m} \times 320\ \mu\text{m}$, achieving a 48% radiation efficiency.

High-Order Harmonic Strategy

To simplify the description of the strategy, in this section we present only one single period cell, i.e. $\Lambda_a = \Lambda_b$ in Fig. 1(b).

The Perturbation Method (PM) was originally formulated by Tamir et al. to provide an efficient semi-analytical framework for extracting the radiation parameters of periodic dielectric waveguides [8]. This approach simplifies the analysis of laminar structures by treating a periodic permittivity layer as a weak perturbation of an otherwise homogeneous waveguide of infinite extent. On the SOI platform, the PM has previously been explored to optimize the heights of the silicon core and buried oxide (BOX) layers to enhance radiation efficiency [9]. However, this strategy has historically been considered less accurate for applications like standard fiber grating couplers, where the assumptions of weak perturbation and infinite extent often fail to hold [10]. For the design of very large antennas in the SOI platform, it is the first time, to the best of the authors' knowledge, that the PM is considered.

We adopt a shallow partial etching strategy where the etch depth H_{etch} is much smaller than

the total core height H_{core} ($H_{\text{core}} = H_{\text{film}} + H_{\text{etch}}$) to be under weak-perturbation constraints to apply the PM. We furthermore reduce the antenna to the two-dimensional (2-D) model illustrated in Fig. 1(b), assuming a standard fabrication stack, with refractive indices $n_{\text{Si}} = 3.476$ and $n_{\text{SiO}_2} = 1.444$ at an operating wavelength $\lambda = 1.55 \mu\text{m}$. Then, the dielectric permittivity of the periodic layer of height H_{etch} is decomposed as

$$\varepsilon(x, z) = \varepsilon_h(z) + \varepsilon_p(x, z), \quad (1)$$

where $\varepsilon_h(z)$ is the volumetric mean of the dielectric permittivity in each layer and $\varepsilon_p(x, z)$ is the periodic perturbation of period Λ . Since ε_p is periodic in x , it can be expanded in Fourier Series (FS) as

$$\varepsilon_p(x, z) = \sum_{m=-\infty}^{\infty} \varepsilon^{(m)}(z) e^{j \frac{2\pi}{\Lambda} m x}, \quad (2)$$

which means that the original structure can be decomposed into a set of independent sub-structures, each associated with one harmonic $\varepsilon^{(m)}$. The zeroth-order term $\varepsilon^{(0)}$ yields a homogeneous multilayer waveguide, from which the Floquet mode effective index n_{FB}^R is computed. The total Floquet field is then reconstructed by superposition of the individual sub-structure fields. As each harmonic is computed independently, it is possible to reduce computational cost by solving only those harmonics which meet the radiation condition $|n_x^{(m)}| < n_{\text{air}}$, with

$$n_x^{(m)} = n_{\text{air}} \sin(\theta^{(m)}) = n_{\text{FB}}^R + \frac{\lambda}{\Lambda} m, \quad (3)$$

where $\theta^{(m)}$ is the radiation angle and λ is the free-space wavelength.

A key result of the PM is that the radiation strength of the antenna scales directly with the weight of the corresponding harmonic in the FS of the dielectric perturbation, $\varepsilon^{(m)}$ [8]. Traditionally, designs exploit the first harmonic ($m = -1$) for maximum power; however, in addition to the shallow etch geometry, we propose using higher-order harmonics to intrinsically achieve even lower radiation strengths. In the following we present an example of this technique applied to an antenna that radiates with the third harmonic ($m = -3$).

Radiating with higher orders requires a larger period Λ , which implies that multiple harmonics may satisfy the grating equation (Eq. 3) simultaneously, producing unwanted sidelobes. To reduce the impact of sidelobes, we employ PM to leverage designs where the duty cycle (DC) of the periodic layer is 50%, which theoretically nullifies all even-order harmonics. As illustrated in the FS of a square perturbation in Fig. 2(a), these even-order sidelobes ideally carry no power even if they fall within the radiation window, i.e. they meet Eq. 3.

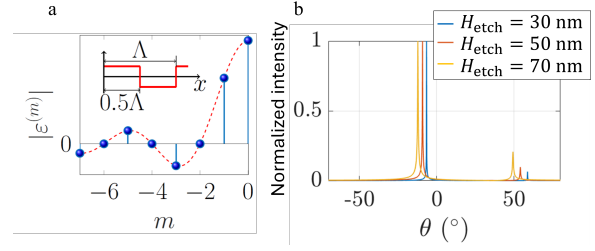


Fig. 2: (a) Fourier Series of a square perturbation. (b) Radiation pattern for TE polarized gratings with $\Lambda = 1.6 \mu\text{m}$ and different etch depths.

This approach is validated by the far-field radiation patterns in Fig. 2(b) for TE polarized gratings with $\Lambda = 1.6 \mu\text{m}$. These far-field patterns have been computed with our in-house software FEXEN [11] for different etching depths of $H_{\text{etch}} = 30, 50$, and 70 nm . For shallow etch depths of 30 nm and 50 nm , the sidelobe level rejection (SLLR) remains over 10 dB , confirming the FS-based suppression. However, at a 70 nm etch depth, the SLLR goes down to approximately 7 dB , indicating that the perturbational approximation begins to fail outside the weak-perturbation regime. Finally, Tab. 1 demonstrates that using a higher period ($\Lambda = 1.6 \mu\text{m}$) such that it radiates with the third harmonic $m = -3$, reduces the radiation strength by an order of magnitude compared to a smaller period ($\Lambda = 0.52 \mu\text{m}$), that radiates with the first harmonic $m = -1$.

Tab. 1: Radiation strength α (Np/mm) vs. H_{etch} (nm) for the first ($m = -1$, $\Lambda = 0.52 \mu\text{m}$) and third ($m = -3$, $\Lambda = 1.6 \mu\text{m}$) harmonics, DC = 50%.

H_{etch} (nm)	α for $\Lambda = 0.52 \mu\text{m}$	α for $\Lambda = 1.6 \mu\text{m}$	α Reduction
30	27.9	3.1	$\times 9$
50	81	6.2	$\times 13$
70	144.1	14.9	$\times 9.7$

Design of a large vertical radiator

From the study presented in the previous section, we conclude that employing high order harmonics in combination with a 30 nm etch depth, could facilitate the synthesis of low radiation strengths.

To satisfy the vertical radiation condition, we consider two concatenated sub-periods Λ_a and Λ_b , which form a composite period $\Lambda = \Lambda_a + \Lambda_b$. As illustrated in the inset of Fig. 3(a), both sub-periods have a DC of 50%. While concatenating two periods was recently proposed in the literature using sub-wavelength (non-radiating) segments [3], our approach differs fundamentally: here, both sub-periods Λ_a and Λ_b are individually radiating periods when employed standalone, so the complete period is radiating with a higher order harmonic, specifically the sixth harmonic $m = -6$.

The use of this dual-period structure allows us to increase the return losses (RL) and deliver a high-performance large vertical antenna. To identify

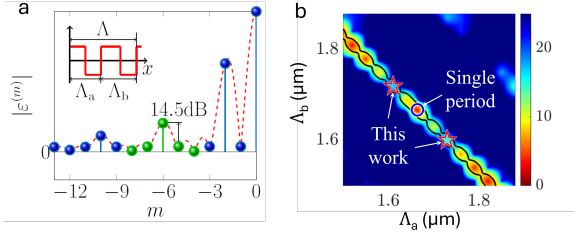


Fig. 3: (a) FS of the selected combination of $\Lambda_a = 1.6 \mu\text{m}$ and $\Lambda_b = 1.73 \mu\text{m}$. Radiating harmonics are shown in green. (b) RL (dB) with sweeps in Λ_a and Λ_b . Region inside black lines indicates vertical radiation.

the optimal combination of these radiating sub-periods, we evaluated the return losses across different combinations of Λ_a and Λ_b in Fig. 3(b) with FEXEN. As shown in the RL map in Fig. 3(b), the diagonal zone enclosed by black curves represents the combinations where vertical radiation is available. When $\Lambda_a = \Lambda_b = 1.67 \mu\text{m}$, the double period is equivalent to a single period perturbation, and therefore, vertical radiation results in high backward reflections with $RL = 0.2 \text{ dB}$ (blue circle in Fig. 3(b)). Within this region, we selected the combination of $\Lambda_a = 1.6 \mu\text{m}$ and $\Lambda_b = 1.73 \mu\text{m}$, marked with a red star in the figure. The reciprocal case is also valid due to the problem symmetry, so it is also marked with a red star.

This design maximizes the return loss with $RL = 13 \text{ dB}$ and achieves a radiation strength of $\alpha = 4.2 \text{ Np/mm}$. This results in an equivalent longitudinal MFD of $320 \mu\text{m}$, which corresponds to a divergence angle of 0.3° . The SLLR reaches 13.8 dB and the radiation efficiency is 52% . Furthermore, the FS of the designed perturbation reveals that the power difference between the sixth-order harmonic and its adjacent orders is 14.5 dB , as shown in Fig. 3(a). This directly explains the computed SLLR of 13.8 dB , showing close agreement with theoretical predictions.

Experimental Results

The proposed antenna was fabricated using a standard silicon-on-insulator manufacturing flow with custom shallow etching provided by the Applied Nanotools foundry. The antenna is driven by a feeder based on a parabolic reflector [12] (Fig. 1(a)). This component expands the fundamental TE mode from a 500 nm -wide waveguide into a transverse Gaussian beam with a MFD of $200 \mu\text{m}$.

The measured far-field radiation pattern is shown in Fig. 4. The full 2-D pattern, centred at $\theta = 0^\circ$, $\phi = 0^\circ$ as expected, is presented in Fig. 4(a). The antenna radiation efficiency is 48% , which is very close to the simulated result of 52% . Based on the simulation of the structure taking into account fabrication errors measured from the SEM, the expected radiation angle at the operating wavelength is $\theta \approx 0.25^\circ$. This is experimentally indistinguishable from perfectly vertical emission

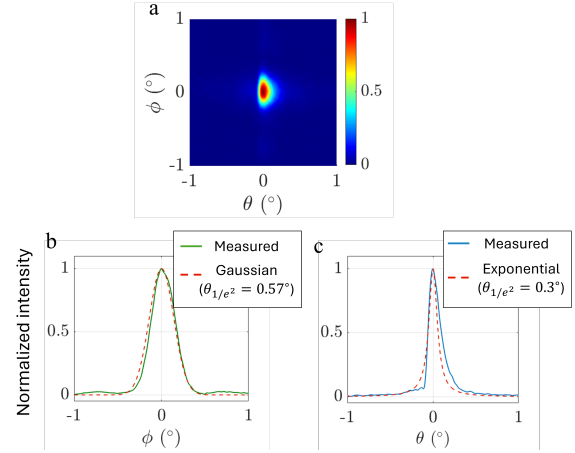


Fig. 4: Measured radiation pattern (a) Full 2-D pattern. (b) Transverse cut. (c) Longitudinal cut. Measured in solid line and analytical Fourier Transform in dashed line.

due to the $\pm 2^\circ$ measurement uncertainty of our setup.

Fig. 4(b) shows the close agreement between the measured transverse cut (in solid line) and the analytical Fourier transform of a $200 \mu\text{m}$ Gaussian (in dashed line). The measured longitudinal cut is shown in Fig. 4(c) (solid line), which is compared to the corresponding radiation pattern of a theoretical exponential decay with $\alpha = 4.2 \text{ Np/mm}$ (dashed line). The measured divergence angle is 0.32° nearly matching the predicted theoretical value of 0.3° .

The strong agreement between our simulations and measurements confirms that our high harmonic order design strategy results in more resilient designs against electron-beam stitching errors, which are typically found in long periodic structures [3]. This is a direct consequence of the fact that using high harmonics results in very long periods, which are less sensitive to absolute pitch errors throughout the structure.

Conclusions

We have presented a design framework based on the classical perturbational analysis to design a perfectly vertical emission large-aperture antenna by leveraging higher order harmonics of a double period grating. We have fabricated the antenna confirming the high-performance radiating pattern of MFD of $200 \mu\text{m} \times 320 \mu\text{m}$, which is the first large-aperture antenna with vertical emission on the SOI platform, and exceeds the beam sizes of results presented on the silicon photonic platforms, to the best of the authors' knowledge. This work therefore paves the way for vertical emission and low strength required in applications such as free space optics and atomic spectroscopy.

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